

The Cell Cycle

- **Definition:** The cell cycle is the sequence of events by which a cell duplicates its genome, synthesizes other cell constituents, and eventually divides into two daughter cells.
- **Purpose:** It is a fundamental process for growth and reproduction in all living organisms.
- **Coordination:** Cell division, DNA replication, and cell growth must occur in a coordinated way to ensure correct division and the formation of progeny cells with intact genomes.
- **Duration:** The duration of the cell cycle varies:
 - Human cells in culture typically divide approximately every 24 hours.
 - Yeast can complete a cell cycle in about 90 minutes.
 - The duration also varies between different organisms and cell types.

Phases of the Cell Cycle

The cell cycle is broadly divided into two basic phases:

1. **Interphase:** The phase between two successive M phases.
2. **M Phase (Mitosis phase):** The phase where actual cell division or mitosis occurs.

- **Duration Distribution:** In a human cell's 24-hour cycle, the M Phase lasts only about an hour, while interphase lasts more than 95% of the cycle duration.

1. Interphase

- **Description:** Although sometimes called the "resting phase," interphase is a period of intense activity where the cell prepares for division by undergoing both cell growth and DNA replication in an orderly manner.
- **Sub-phases:** Interphase is further divided into three phases:
 - **G1 phase (Gap 1):**
 - Corresponds to the interval between mitosis and the initiation of DNA replication.

- The cell is metabolically active and continuously grows.
- Does not replicate its DNA during this phase.
- Most organelle duplication occurs during this phase.
- **S phase (Synthesis):**
 - Marks the period during which DNA synthesis or replication takes place.
 - During this time, the amount of DNA per cell doubles; if the initial DNA content is $2C$, it increases to $4C$.
 - Crucially, there is no increase in chromosome number; if a cell has a diploid ($2n$) number of chromosomes at G_1 , it remains $2n$ after S phase.
 - In animal cells, centriole duplication occurs in the cytoplasm during S phase, alongside DNA replication in the nucleus.
- **G₂ phase (Gap 2):**
 - Proteins are synthesized in preparation for mitosis.
 - Cell growth continues during this phase.
- **Quiescent Stage (G_0):**
 - Some adult animal cells, such as heart cells, do not exhibit division.
 - Many other cells divide only occasionally, for instance, to replace lost cells due to injury or cell death.
 - These cells exit the G_1 phase and enter an inactive stage called the quiescent stage (G_0).
 - Cells in G_0 remain metabolically active but no longer proliferate unless there is a specific requirement from the organism.

II. M Phase (Mitosis Phase)

- **Description:** This is the most dramatic period of the cell cycle, involving a major reorganization of virtually all cell components.
- **Equational Division:** Mitosis is also called equational division because the number of chromosomes in the parent and

progeny cells remains the same.

• **Stages: Mitosis involves two main parts:**

1. Karyokinesis: Nuclear division, specifically the segregation of duplicated chromosomes into daughter nuclei.
2. Cytokinesis: Division of the cytoplasm.

Karyokinesis (Nuclear Division)

Karyokinesis is a progressive process and is divided into four stages:

1. Prophase:

- **Initiation:** Follows the S and G₂ phases of interphase.
- **Chromosomal Condensation:** Marked by the initiation of condensation of chromosomal material, making the chromosomes compact mitotic chromosomes.
- **Sister Chromatids:** Chromosomes are seen to be composed of two chromatids attached at the centromere.
- **Centrosome Movement:** The centrosome (duplicated during S phase) begins to move towards opposite poles of the cell.
- **Mitotic Apparatus:** Each centrosome radiates microtubules called asters. The two asters, along with spindle fibers, form the mitotic apparatus.
- **Disappearance of Organelles:** By the end of prophase, structures like Golgi complexes, endoplasmic reticulum, nucleolus, and the nuclear envelope disappear when viewed under a microscope.

2. Metaphase:

- **Nuclear Envelope Disintegration:** The complete disintegration of the nuclear envelope marks the start of this phase, spreading chromosomes through the cell's cytoplasm.
- **Chromosome Condensation:** Chromosome condensation is completed, making them clearly observable under the microscope, allowing for easy study of their morphology.
- **Metaphase Chromosome Structure:** A metaphase chromosome

consists of two sister chromatids held together by the centromere.

- **Kinetochore:** Small disc-shaped structures on the surface of centromeres called kinetochores serve as attachment sites for spindle fibers.

- **Alignment at Metaphase Plate:** All chromosomes come to lie at the equator of the cell, forming the metaphase plate.

- One chromatid of each chromosome connects to spindle fibers from one pole.

- Its sister chromatid connects to spindle fibers from the opposite pole.

3. Anaphase:

- **Centromere Splitting:** At the onset, each chromosome on the metaphase plate splits simultaneously at the centromere.

- **Chromatid Separation:** The two daughter chromatids (now considered daughter chromosomes) begin migrating towards opposite poles of the cell.

- **Movement Mechanism:** As chromosomes move, their centromeres lead towards the pole, with the arms trailing behind.

4. Telophase:

- **Decondensation:** Chromosomes reaching their respective poles decondense and lose their individuality, becoming a mass of chromatin material at each pole.

- **Nuclear Envelope Formation:** A nuclear envelope develops around the chromosome clusters at each pole, forming two daughter nuclei.

- **Reformation of Organelles:** The nucleolus, Golgi complex, and ER reform.

Cytokinesis (Cytoplasmic Division)

- **Completion of Cell Division:** Follows karyokinesis, dividing the cell's cytoplasm into two daughter cells.

- In Animal Cells: Achieved by the appearance of a furrow in the plasma membrane, which gradually deepens and divides the cytoplasm.
- In Plant Cells: Due to the inextensible cell wall, cytokinesis occurs differently. Wall formation starts in the center of the cell and grows outward to meet existing lateral walls. This new wall begins with a cell-plate, which represents the middle lamella between adjacent cells.
- **Organelle Distribution:** Organelles like mitochondria and plastids are distributed between the two daughter cells during cytoplasmic division.
- **Syncytium:** In some organisms, karyokinesis is not followed by cytokinesis, leading to a multinucleate condition called a syncytium (e.g., liquid endosperm in coconut).

Significance of Mitosis

- **Growth:** Responsible for the growth of multicellular organisms.
- **Cell Repair/Replacement:** Essential for cell repair and replacement; for example, cells of the upper epidermis, gut lining, and blood cells are constantly replaced.
- **Restoration of Nucleo-cytoplasmic Ratio:** Cell growth can disturb the ratio between the nucleus and cytoplasm, making cell division essential to restore this ratio.
- **Chromosome Number Conservation:** Usually results in diploid daughter cells with identical genetic complement to the parent cell.
- **Haploid Cell Division:** While typically restricted to diploid somatic cells in animals, mitotic cell division can occur in haploid cells in some lower plants and social insects (e.g., male honey bees). Plants can show mitotic divisions in both haploid and diploid cells.

Meiosis

- **Purpose:** A specialized kind of cell division that reduces the chromosome number by half to produce haploid daughter cells (gametes).
- **Gametogenesis:** Meiosis occurs during gametogenesis in plants and animals, leading to the formation of haploid gametes.
- **Chromosome Number Restoration:** Ensures the production of a haploid phase in sexually reproducing organisms, and fertilization restores the diploid phase.
- **Genetic Variability:** Increases genetic variability in the population from one generation to the next, which is crucial for evolution.
- **Key Features:**
 - Involves two sequential cycles of nuclear and cell division (Meiosis I and Meiosis II).
 - Involves only a single cycle of DNA replication, which occurs before Meiosis I, during the S phase, producing identical sister chromatids.
 - Features the pairing of homologous chromosomes and recombination (crossing over) between non-sister chromatids.
 - Results in the formation of four haploid cells at the end of Meiosis II.

1. Meiosis I

- **Reductional Division:** Homologous chromosomes separate, reducing the chromosome number by half.
- **Prophase I:**
 - Typically longer and more complex than mitotic prophase.
 - Subdivided into five phases based on chromosomal behavior:
 1. **Leptotene:** Chromosomes become gradually visible under a light microscope, and compaction continues.
 2. **Zygotene:** Chromosomes start pairing together (synapsis). The paired chromosomes are called homologous chromosomes. A complex structure called the synaptonemal complex forms,

resulting in a complex of synapsed homologous chromosomes known as a bivalent or a tetrad (composed of four chromatids).

3. Pachytene: The four chromatids of each bivalent become distinct and clearly appear as tetrads. This stage is characterized by the appearance of recombination nodules, the sites where crossing over occurs between non-sister chromatids of homologous chromosomes. Crossing over is the exchange of genetic material and is an enzyme-mediated process involving recombinase. Recombination is completed by the end of pachytene.

4. Diplotene: Recognized by the dissolution of the synaptonemal complex and the tendency of recombined homologous chromosomes to separate, except at the sites of crossovers. These X-shaped structures are called chiasmata. (In some vertebrate oocytes, this stage can last for months or years).

5. Diakinesis: Marked by the terminalisation of chiasmata. Chromosomes are fully condensed, and the meiotic spindle is assembled. By the end, the nucleolus disappears, and the nuclear envelope breaks down. It represents the transition to metaphase.

• **Metaphase I:**

- The bivalent chromosomes align on the equatorial plate.
- Microtubules from opposite poles attach to the kinetochore of homologous chromosomes (one kinetochore per homologous chromosome, not per chromatid as in mitosis).

• **Anaphase I:**

- Homologous chromosomes separate and move towards opposite poles.
- Crucially, sister chromatids remain associated at their centromeres. Each pole receives half the chromosome number

of the parent cell.

• **Telophase I:**

- The nuclear membrane and nucleolus reappear.
- Cytokinesis follows, resulting in a dyad of cells (two haploid cells).
- Chromosomes may undergo some dispersion but generally do not reach the extremely extended state of the interphase nucleus.

• **Interkinesis:**

- The stage between the two meiotic divisions.
- Generally short-lived.
- No replication of DNA occurs during interkinesis.

II. Meiosis II

• **Resemblance to Mitosis:** Meiosis II is initiated immediately after cytokinesis (Meiosis I) and resembles a normal mitosis. It is an equational division.

• **Prophase II:**

- The nuclear membrane disappears by the end of this phase.
- Chromosomes become compact again.

• **Metaphase II:**

- Chromosomes align at the equator.
- Microtubules from opposite poles attach to the kinetochores of sister chromatids.

• **Anaphase II:**

- Begins with the simultaneous splitting of the centromere of each chromosome (which held sister chromatids together).
- Sister chromatids, now considered individual chromosomes, move towards opposite poles of the cell due to the shortening of microtubules.

• **Telophase II:**

- The two groups of chromosomes at each pole are enclosed by a nuclear envelope.

- Cytokinesis follows, leading to the formation of a tetrad of cells (four haploid daughter cells). Each cell is haploid and contains a single set of chromosomes.

Significance of Meiosis

- Conservation of Chromosome Number: It is the mechanism by which the specific chromosome number of each species is conserved across generations in sexually reproducing organisms. Although it halves the chromosome number during gamete formation, fertilization restores it.
- Increased Genetic Variability: Meiosis significantly increases the genetic variability within a population due to crossing over and independent assortment of homologous chromosomes. These variations are very important for the process of evolution